Theoretical Formation & Evolution of CBCs

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BHs found in two regimes by electromagnetic (EM) observations: stellar-mass and supermassive BHs.

- The stellar BHs have been confirmed with direct detections of their mergers by GWs (see LVK catalogs).
- The existence of BHs of intermediate mass (IMBHs) are uncertain, but EM obs strongly suggest they exist (Mezcua, 2017; Barrows *et al.*, 2019), and you can consider the post-merger of GW190521 an IMBH.
- NOTE: the boundaries b/w the BH mass regimes are not rigorous, which reflects the fact that GR is a scale-free theory and black hole mass is a free parameter of black hole solutions, ie, pure GR gives us no hints for this!
- Supermassive BHs are interesting, as they're intimately involved with the cosmic evolution of galaxies, and hence the fate of the Universe, but let's focus on stellar mass BHs because LIGO is sensitive to their mergers. These BHs are the result of stellar evolution.

Some basics of single stellar evolution:

- A star is not a ball of fire: it's a highly dense and magnetized *plasma*.
- See, e.g., S. Chandrasekhar's book, An Intro to the Study of Stellar Structure, for the equations that evolve a star's temperature, density, and luminosity as a function of radius and time. These build on centuries of work of models of polytropic fluids (ie see the Lame-Emden equation) by incorporating nuclear physics.
- Stars form from gravitational collapse of molecular gas clouds or fragmented accretion disks, etc. The impending gravitational collapse causes very high temperatures and densities in the collapsing cloud until nuclear fusion is achieved which works to balance the collapse.
- A main sequence star burns hydrogen into helium at its center via two main types of nuclear fusion (which is an exothermic reaction): proton-proton chain reaction, and the carbon-nitrogen-oxygen cycle. This is basic stellar nucleosynthesis.
- Thus, a star is a beautiful balancing act between inward gravitational pressure acting to crush everything and outward (mechanical/thermodynamic/etc.) pressures acting to maintain hydrostatic equilibrium.
- QUESTION: how does the lifetime of a star depend on its mass? Directly or inversely proportional?
- The Hertzsprung-Russell diagram has luminosity vs temperature (or equivalently, brightness vs stellar type). It shows either (1) a scatterplot of observed stars, or (2) tracks of stars as they age through stages of evolution.
- As it burns hydrogen into helium, a core begins to develop. The star evolves across various stages of evolution that correspond to the ignition of new sources of nuclear fuel, e.g., the star leaves the main sequence when its luminosity is dominated by helium burning rather than hydrogen burning. Eventually, the star acquires shells of heavy elements around a mostly helium core. This continues to iron b/c iron has peak binding energy.

- Nuclear fusion works by fusing two lighter elements together to obtain a heavier element. This is only achievable in stars up to (and including) iron because the binding energy per nucleon is maximised in nuclei around the iron peak, implying that if you tried to a neutron to an iron nucleus, the resulting nucleus would have a smaller binding energy per nucleon, which is like removing water from a deep well and placing it in a shallower well, which requires input energy.

- Thus, to produce elements heavier than iron requires input energy, i.e., an endothermic reaction called nuclear fission. This occurs when stars die in supernovae explosions, etc.

- When the star runs out of fuel for nuclear fusion, it can no longer support itself against gravitational collapse and thus dies in a supernova. As a result, a compact object forms - its type depends on the mass and composition of the progenitor star (see e.g., Heger, 2003).

- Without nuclear fuel, the fermionic quantum degeneracy pressure supports against gravitational collapse - first electrons (white dwarf) and then neutrons (neutron star) and if massive enough then gravitational collapse is unstoppable and you get a black hole (or something more exotic?).

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- There are various incarnations of stellar evolution models that offer a range of sophistication and applicability.
  - (Magneto-) hydrodynamical simulations that resolve the fluid dynamics of the gas particles focusing on macro- or micro- physics aspects of thermodynamic and radiative processes.

  - Numerical simulations of various types and scales which offer less detailed resolution than hydro sims but are computationally cheaper.

  - Analytic formulas fitted to the tabular results of numerical simulations. These are computationally cheapest, but rely on simplified parameterizations of many important (and uncertain) processes.

Theoretical bounds on the masses of compact objects:

- In GR, the maximum mass of a WD is the Chandrasekhar Limit, about 1.4 M_{sol} (depending on stability), and the maximum mass of a NS is the Tolman-Volkoff limit, about 1.5 - 3.0 M_{sol} but it depends on rotation and EOS etc.

- Historically, stellar evolution and gravitational collapse theory predicted a minimum BH mass of $\sim 5\ M_{sol}$, giving rise to the expection of a "lower mass gap" b/w NSs and BHs. However, this is very uncertain since it depends on the uncertainties of supernova explosion mechanisms… even if there was a lower mass gap it may be occupied by the remnants of compact binary mergers, so detecting the gap would be challenging. There's also an "upper mass gap" for BHs at $\sim 50\ -150\ M_{sol}$ due to pair instability supernovae, but the bounds are very controversial.

- The maximum mass of a BH is given theoretically from cosmological simulations of accreting supermassive BHs to be $\sim 10^{11}\ M_{sol}$ (Natarajan & Treister, 2009).

- Lets focus on BHs (but most of what follows also applies to WDs and NSs, just scaled by mass), implying that we're interested in the evolution of high mass stars, ie., $M_{ZAMS}\ \gtrsim 20\ M_{sol}$ for sub-solar metallicities.

Most high mass stars that form BHs are known to exist with a companion (e.g., Sana *et al.*, 2012).

- The first stellar mass BHs were found in binaries, and were observed from their accretion as so-called "X-ray binaries" where a BH is accreting gas from its stellar binary companion.

- Although these binaries show us a very interesting phase of stellar binary evolution, recent studies show that they most likely compose a different subpopulation then the binaries that LIGO observes (e.g., Fishbach &

Kalogera, 2022), ie, X-ray binaries are generally at smaller redshift and in higher metallicity environments.

Let's focus on sources of gravitational waves (GWs):

1. GW chirp from binary black hole (BBH) merger:
   → We require binaries that merge within the age of the Universe (b/c otherwise, how will we catch the GW?), implying that the BBH must form with sufficiently small binary separation. Also, the BBH must not be too massive or it'll merge at a frequency above the LIGO detection band.

   It's like chicken or egg: does the BH or the binary form first?... This leads to two main possible formation channels:

   i) Cluster/dynamical formation channel: individual BHs form first, then they form binaries through dynamical interactions (ie., gravitational scattering). This occurs in dense stellar clusters, e.g.'s, open, young, globular, nuclear clusters. This is modeled by numerical N-body simulations. See e.g., Benacquista & Downing, 2013

   ii) Isolated formation channel: binary star forms first, then evolves through the kindred fate of stellar binary evolution (ie., many relevant astrophysical processes). This occurs in the galactic field. This is modeled by either (a) detailed numerical sims of a small number of stellar binaries via a code such as MESA, or (b) analytic parameterizations of a large number of stellar binaries via "rapid population synthesis" codes. See e.g., Postnov & Yungelson, 2014

   Note: the single stellar evolution models mentioned earlier are used in models of both of these BBH formation scenarios, implying that uncertainties in single stellar evolution propagate to BBH evolution!

These two formation channels provide different predictions for the properties of BBHs:

i) Cluster:
- multiple generations of mergers, some may be more likely to survive than others
- these BBHS generally have nonzero eccentricity at merger because they are driven to merger quicky via gravitational runaway and thus don't have sufficient time to circularize from GW emission
- the typical values of the BH dimensionless spins for these BBHs is $\chi \simeq 0.7$
- the BH spin orientations are expected to be mostly isotropic due to repeated scattering

ii) Isolated:
- generally treated as evolving in isolation, or can be embedded in AGN accretion disk.
- these BBHs generally have zero eccentricity because they form at wide separations, i.e., $r/M \simeq 10^5$, and have sufficient time to circularize due to GW emission before merger.
- binary stellar evolution involves many astrophysically uncertain processes, and thus the spin magnitudes and orientations of isolated BBHs are uncertain. Traditionally, isolated BBHs are thought to have small spin magnitudes and spin-orbit misalignments. My PhD was about challenging this orthodoxy!

Some astrophysical processes relevant for mass and spin evolution of isolated BBHs:

- **Stellar winds** from a hot, luminous star's surface are best explained by the line-driven model in which the wind mass-loss rate is an increasing function of stellar mass and metallicity. The winds remove mass and angular momentum from the star. See e.g., the works by J. G. L. M. Lamers.

- Stellar **angular momentum transport within the stellar interior determines the initial spin of the resultant BH**. If stellar ang. mom. transport is efficient then the stellar core will be depleted of spin producing small BH spin; conversely, if the transport is inefficient then the BH will have high spin. This is a **HUGE uncertainty in stellar evolution models** and could be a promising mechanism to obtain highly spinning isolated BHs.

- Mass transfer processes between the two binary components dominate the evolution of the binary's ang. mom. at large separations before GW emission becomes dominant at smaller separations. Mass transfer is complicated, but generally follows from Roche lobe overflow (RLOF), ie when the star can no longer contain itself in a binary gravitational potential:

  Two main examples of mass transfer are:

  - Dynamically **unstable** mass transfer (also called **common envelope evolution**), where the donor star loses its entire envelope and the companion cannot accrete fast enough so the binary is surrounded by the donor's envelope which either results in a premature stellar merger or in expulsion of the envelope by drastically reducing the binary separation (ie energy is transferred from the orbit to the envelope). This is **crucial for producing BBHs** with sufficiently small separations **that merge within the age of the Universe**, where the timescale to merge from GW emission alone goes as $t \sim r^4$.

  - Dynamically **stable** mass transfer which allows for **sustained accretion episodes** that increase the mass and spin of the accretor. This is another promising mechanism to obtain highly spinning BHs.

- When the BH forms it will experience a **recoil velocity** due to conservation of linear momentum, these recoil velocities are called **natal kicks** and are thought to be associated with supernovae. Traditionally associated with NSs, it's become increasingly expected that BHs probably receive kicks as well at formation. Natal kick are the **main source of spin-orbit misalignments for isolated BBHs**, and are thus crucial for producing

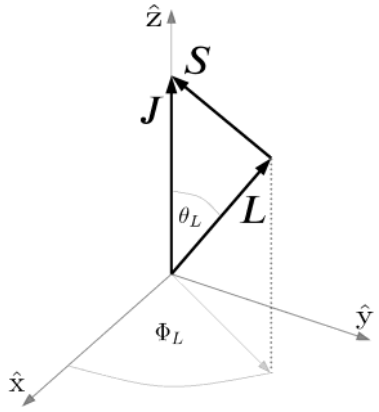
precessing isolated BBHs.

- Tidal interactions can act on a star in the binary, which work to exchange kinetic energy and angular momentum b/w the rotation of the binary components and their orbital motion. This tends to **align the star's spin** with the orbital ang. mom., and to **circularize the orbit** (ie, eccentricity goes to zero).

Some notes on 2-body general-relativistic spin precession:

The different predictions for the spin orientations of BBHs from different formation channels translates into different predictions for the relativistic spin precession of the BBH during inspiral and merger. This spin precession modulates the GW in frequency, phase, and amplitude. For example, amplitude modulation in the BBH strain at 2PN order,

We can describe the spin precession of an inspiraling binary in post-Newtonian theory with a timescale hierarchy $t_{\rm orb} << t_{pre} << t_{rr}$ which simplifies the dynamics so we can decompose the evolution of L into two motions: azimuthal *precession* of L on a cone about the total ang mom J, and polar nutation of L.



That is, using conservation of total ang mom, ie J = S + L,

- **Regular** precession occurs when S is time independent so the opening angle of L w.r.t to J is also constant in time, allowing L to rotate about J in a cone with constant opening angle.

- **Generic** precession occurs when S depends on time, so the opening angle of L also changes in time causing L to nutate (ie, it raises and lowers like a draw bridge).

-> the simpler formation prediction for **cluster** channel, ie moderate spin magnitudes and isotropic spin orientations, results in **generic** precession because there will be precession of L and nutation of L.

-> the isolated channel has a lot more astrophysical possibilities (uncertainties), so the spin precession predictions depend on various astrophyscial processes, see Steinle & Kesden, 2022. Generic precession is generally possible from high BH spins via inefficient stellar angular momentum transport.

* See V. Varma's Black Hole Explorer which includes lots of neat videos/animations!

Importantly, LVK is not yet sensitive enough to constrain the individual spin magnitudes or orientations of BBHs, so instead we have to use effective spin parameters that come from PN theory: $\chi_{eff} = (\chi_1 \cos\theta_1 + \chi_2\, q \cos\theta_2)/(1+q)$ which is a constant of motion on the precession timescale and encodes the aligned spin-component information, and χ_p which is slightly controversial and encodes the orbital-plane spin-component

information.

*** However, as LVK sensitivity improves, and with the advent of 3G detectors, we will cast these effective parameters aside in favor of better parameterizations, see e.g. Gangardt *et al.*, 2022. ***

*** Currently, LVK data is not sufficient to confidently identify the likely formation scenario of any of the observed BBHs thus far. Also, we do not yet have sufficiently accurate waveforms that simultaneously model nonzero eccentricity and generic spin precession (this is possible in post-Newtonian regime in theory, but is very challenging in practice to make a full inspiral-merger-ringdown waveform).***

2. Continuous GWs from isolated pulsar:
   i) Could form directly from a single intermediate-mass star that dies, or from the merger of two WDs or NSs, or (more exotically) from a BH-NS merger.
   ii) The NS is thought to be born with some spin from gravitational collapse (very uncertain!), depending on its progenitor's mass, spin, and magnetic field, but that spin is expected to be diminished due to magnetic breaking as it ages.
   iii) See A. Buonano (2007) section 7.1 for the GW strain amplitude of continuous GWs from a pulsar - as Lucas showed us, the strain depends on the NS ellipticity. The EM observed energy loss of pulsars cannot be explained by GWs alone!